

Chapter 3
Image Enhancement in the Spatial Domain

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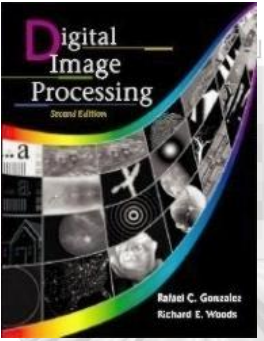
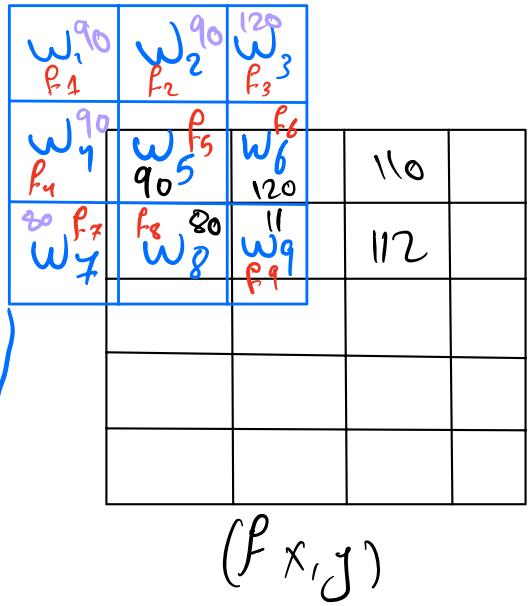


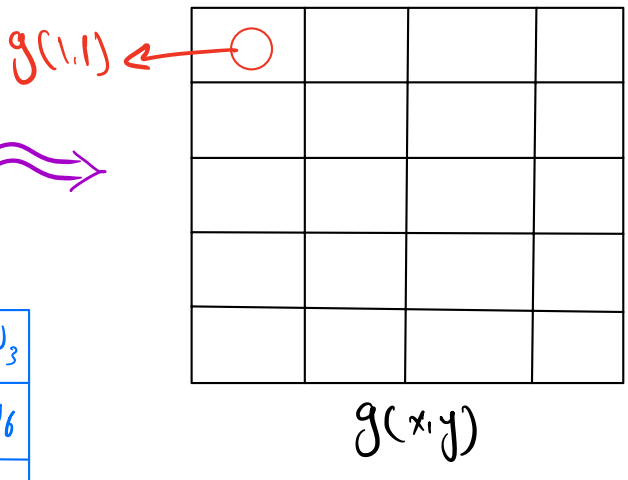
Image Enhancement in the Spatial Domain

- The **spatial domain**:
  - The image plane
  - For a digital image is a Cartesian coordinate system of discrete rows and columns. At the intersection of each row and column is a pixel. Each pixel has a value, which we will call intensity.
- The **frequency domain** :
  - A (2-dimensional) discrete Fourier transform of the spatial domain
  - We will discuss it in chapter 4.
- **Enhancement** :
  - To "improve" the usefulness of an image by using some transformation on the image.
  - Often the improvement is to help make the image "better" looking, such as increasing the intensity or contrast.

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Filtering



w(s,t)

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

Values
* to identify the filter type

اسماء أخرى
mask
kernel
window

* square and odd size (3x3, 5x5, ...)

① Zero padding
→ f_1, f_2, f_3, f_4, f_7 → 0

② replicate
→ $f_1 = f_5$ $f_4 = f_5$
$f_2 = f_5$ $f_7 = f_8$
$f_3 = f_6$

$g(x,y) = f_1 * w_1 + f_2 * w_2 + f_3 * w_3 + \cdots + f_9 * w_9$

① zero padding $g(1,1) = 0*\frac{1}{9} + 0*\frac{1}{9} + 0*\frac{1}{9} + 0*\frac{1}{9} + 90*\frac{1}{9} + 120*\frac{1}{9} + 0*\frac{1}{9} + 80*\frac{1}{9} + 11*\frac{1}{9}$

$= \frac{90+120+80+11}{9} = 33.\overline{4}$

② replicate $g(1,1) = 90*\frac{1}{9} + 90*\frac{1}{9} + 120*\frac{1}{9} + 90*\frac{1}{9} + 90*\frac{1}{9} + 120*\frac{1}{9} + 80*\frac{1}{9} + 80*\frac{1}{9} + 11*\frac{1}{9}$

$= \frac{90+90+120+90+90+120+80+80+11}{9} = 85.\overline{66}$

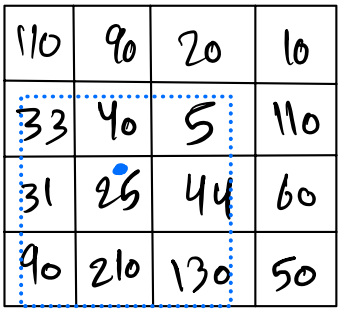
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spatial filters:

① Smoothing:
- Linear → Box (standard) → weighted Av.
- non linear

w_1 $\frac{1}{9}$	w_2 $\frac{1}{9}$	w_3 $\frac{1}{9}$
w_4 $\frac{1}{9}$	$\frac{1}{9}$ w_5	$\frac{1}{9}$ w_6
w_7 $\frac{1}{9}$	$\frac{1}{9}$ w_8	$\frac{1}{9}$ w_9

② Sharpening

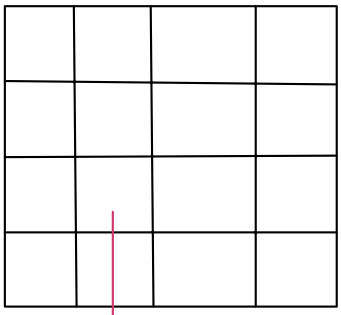


$f(x, y)$

Filtering

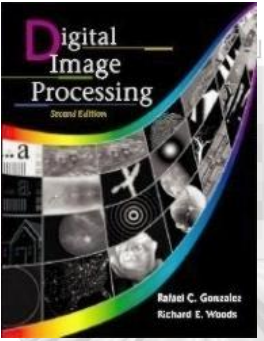
$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$
$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{8}$
$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{16}$

Weighted Avg.



$g(x, y)$

$$= 33 \times \frac{1}{16} + 40 \times \frac{1}{8} + 5 \times \frac{1}{16} + 31 \times \frac{1}{8} + 25 \times \frac{1}{4} + 44 \times \frac{1}{8} + 90 \times \frac{1}{16} + 260 \times \frac{1}{8} + 130 \times \frac{1}{16} = 63$$



Basics of Spatial Filtering

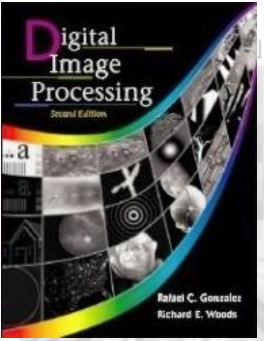
- In spatial filtering (vs. frequency domain filtering), the output image is computed directly by simple calculations on the pixels of the input image.
- Spatial filtering can be either linear or non-linear.
- For each output pixel, some neighborhood of input pixels is used in the computation.
- In general, linear filtering of an image f of size MXN with a filter mask of size mxn is given by

$$g(x,y) = \sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

where a=(m-1)/2 and b=(n-1)/2

- This concept called convolution. Filter masks are sometimes called convolution masks or convolution kernels.

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Basics of Spatial Filtering

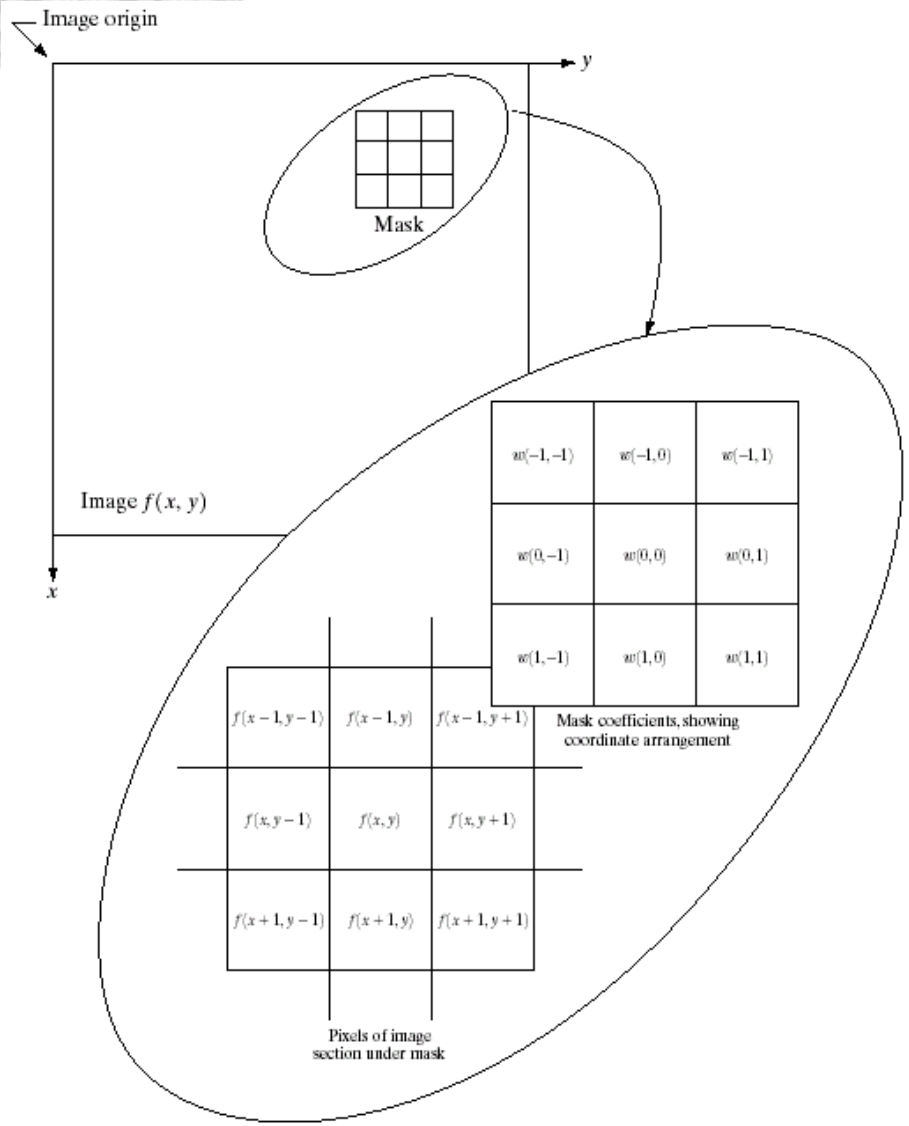
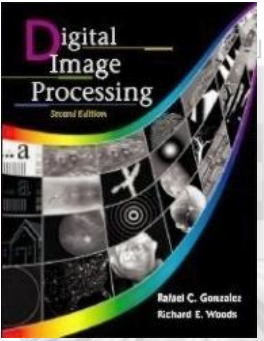


FIGURE 3.32 The mechanics of spatial filtering. The magnified drawing shows a 3×3 mask and the image section directly under it; the image section is shown displaced out from under the mask for ease of readability.

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

FIGURE 3.33 Another representation of a general 3×3 spatial filter mask.

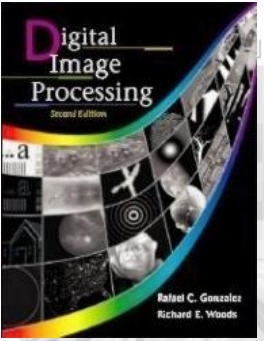
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Basics of Spatial Filtering

- Nonlinear spatial filtering usually uses a neighborhood too, but some other mathematical operations are use. These can include conditional operations (if …, then…), statistical (sorting pixel values in the neighborhood), etc.
- Because the neighborhood includes pixels on all sides of the center pixel, some special procedure must be used along the top, bottom, left and right sides of the image so that the processing does not try to use pixels that do not exist.

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Smoothing Spatial Filters

- Smoothing linear filters
  - Averaging filters (Lowpass filters in Chapter 4))
    - Box filter
    - Weighted average filter

$\frac{1}{9} \times$

1	1	1
1	1	1
1	1	1

Box filter

$\frac{1}{16} \times$

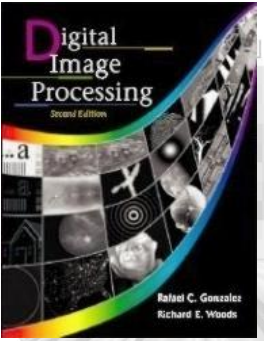
1	2	1
2	4	2
1	2	1

Weighted average

a b

FIGURE 3.34 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to the sum of the values of its coefficients, as is required to compute an average.

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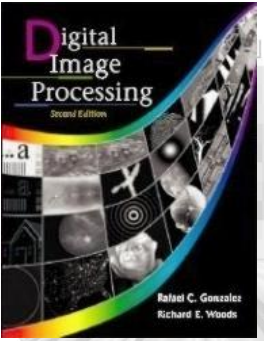
Smoothing Spatial Filters

- The general implementation for filtering an MXN image with a weighted averaging filter of size mxn is given by

$$g(x,y) = \frac{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t) f(x+s, y+t)}{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)}$$

where a=(m-1)/2 and b=(n-1)/2

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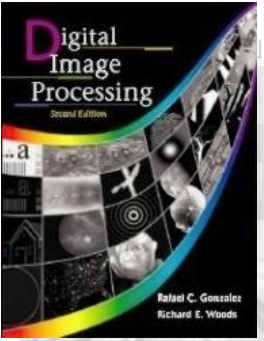


- Smoothing filters are used for blurring and for noise reduction.
- Blurring is used in preprocessing steps, such as removal of small details from an image prior to

(large) object extraction, and bridging of small gaps in lines or curves.

Noise reduction can be accomplished by blurring with a linear filter and also by non-linear filtering.

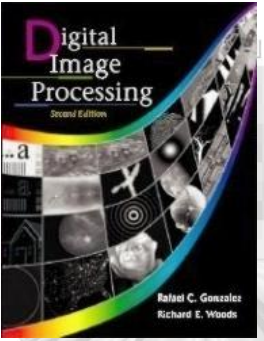
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- By replacing the value of every pixel in an image by the average of the gray levels in the neighborhood defined by the filter mask, this process results in an image with reduced

"sharp" transitions in gray levels.

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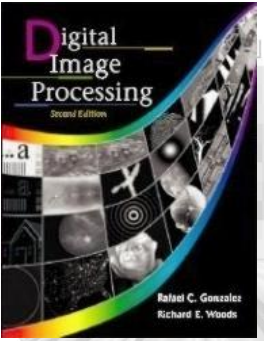


Smoothing Spatial Filters

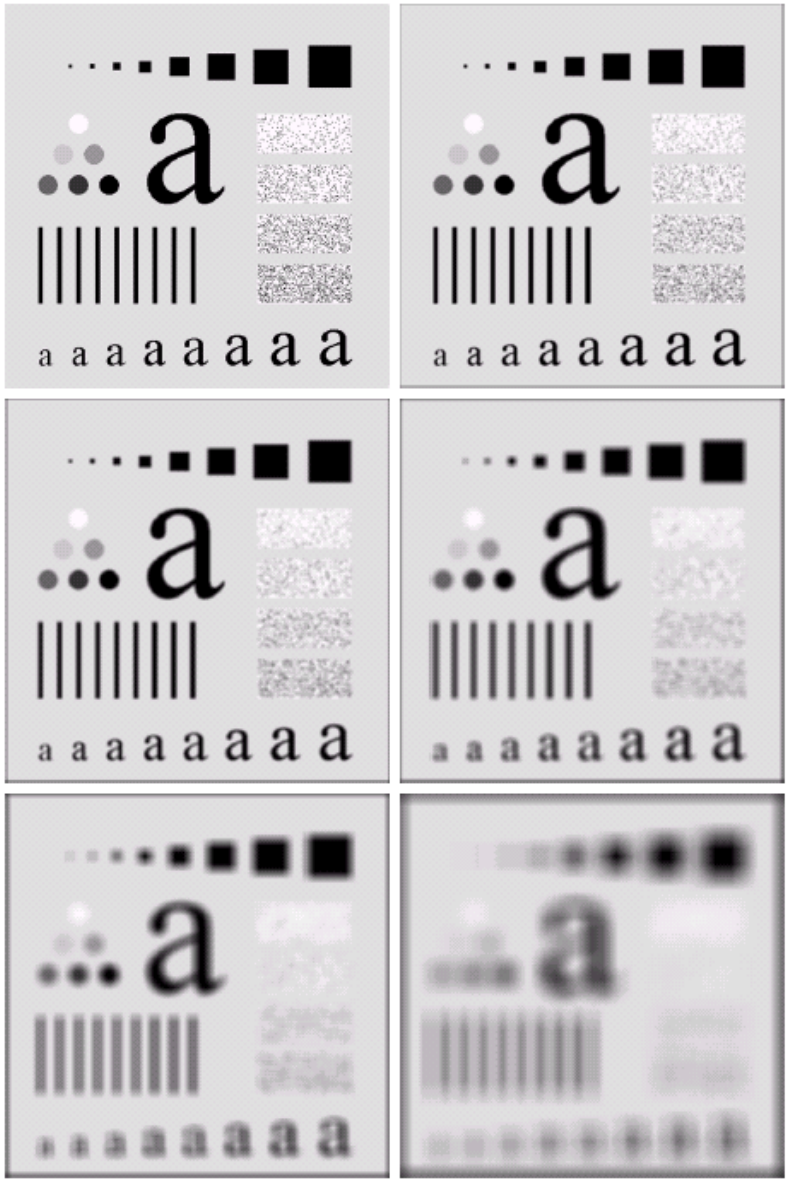
- Smoothing filters are used for blurring and for noise reduction.
- Blurring is used in preprocessing steps, such as removal of small details from an image prior to (large) object extraction, and bridging of small gaps in lines or curves.

  Noise reduction can be accomplished by blurring with a linear filter and also by nonlinear filtering.

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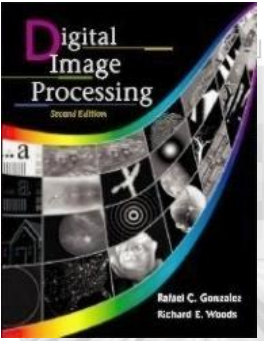
Smoothing Spatial Filters
Image smoothing with masks of various sizes



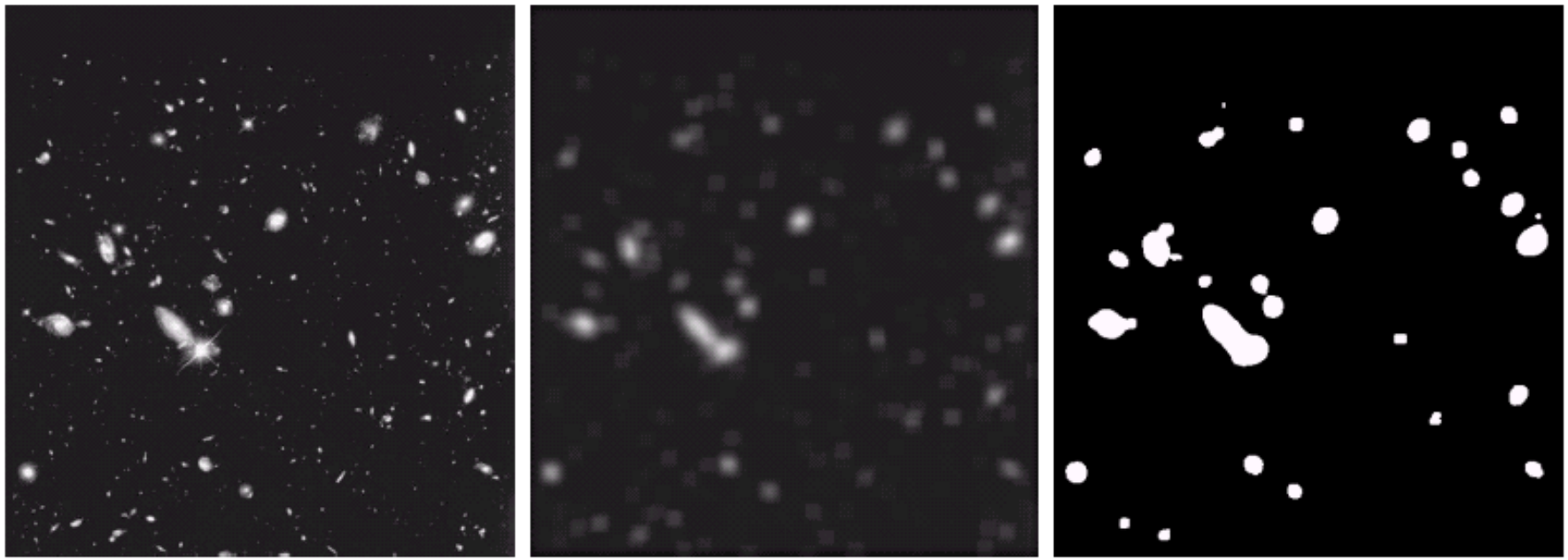
a b
c d
e f

FIGURE 3.35 (a) Original image, of size 500 × 500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $n = 3, 5, 9, 15$, and 35, respectively. The black squares at the top are of sizes 3, 5, 9, 15, 25, 35, 45, and 55 pixels, respectively; their borders are 25 pixels apart. The letters at the bottom range in size from 10 to 24 points, in increments of 2 points; the large letter at the top is 60 points. The vertical bars are 5 pixels wide and 100 pixels high; their separation is 20 pixels. The diameter of the circles is 25 pixels, and their borders are 15 pixels apart; their gray levels range from 0% to 100% black in increments of 20%. The background of the image is 10% black. The noisy rectangles are of size 50 × 120 pixels.

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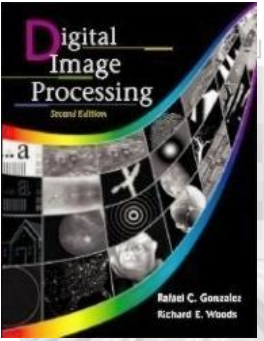
Smoothing Spatial Filters
Another Example



a b c

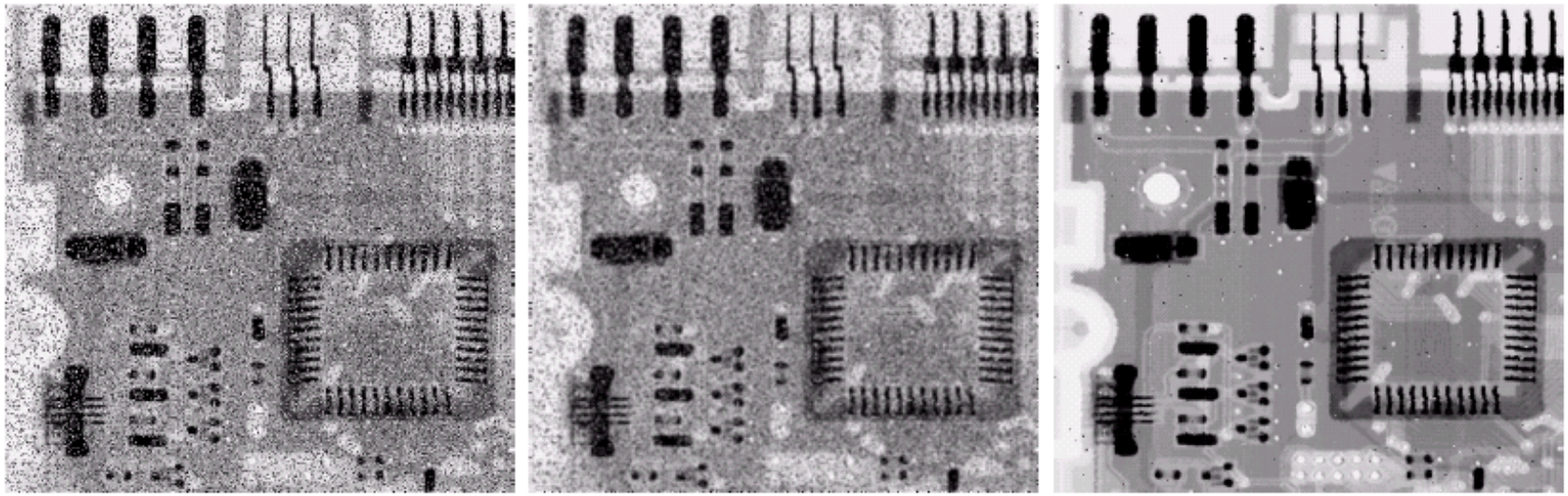
FIGURE 3.36 (a) Image from the Hubble Space Telescope. (b) Image processed by a 15 × 15 averaging mask. (c) Result of thresholding (b). (Original image courtesy of NASA.)

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Order-Statistic Filters

- Order-statistic filters
  - Median filter: to reduce impulse noise (salt-and-pepper noise)



a b c

FIGURE 3.37 (a) X-ray image of circuit board corrupted by salt-and-pepper noise. (b) Noise reduction with a 3 × 3 averaging mask. (c) Noise reduction with a 3 × 3 median filter. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

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non_linear: (statistics)
Median Filter

110	90	20	10	
33	40	5	110	0
31	25	44	60	0
90	210	130	50	0

$f(x,y)$

Filtering

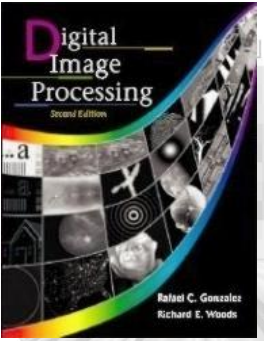
3×3

	40?		44

$g(x,y)$

5, 25, 31, 33, 40, 44, 90, 130, 210

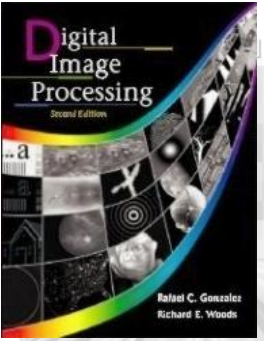
0, 0, 0, 5, 44, 50, 60, 110, 130



Sharpening Spatial Filters

- Sharpening filters are based on computing spatial derivatives of an image.
- The first-order derivative of a one-dimensional function $f(x)$ is $\frac{\partial f}{\partial x} = f(x+1) - f(x)$
- The second-order derivative of a one-dimensional function $f(x)$ is $\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$

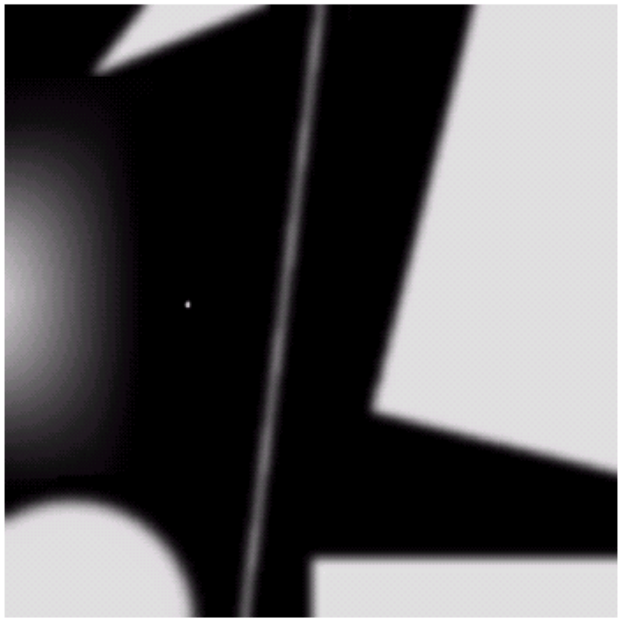
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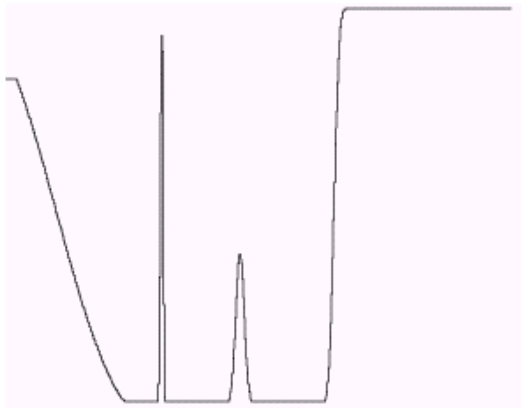


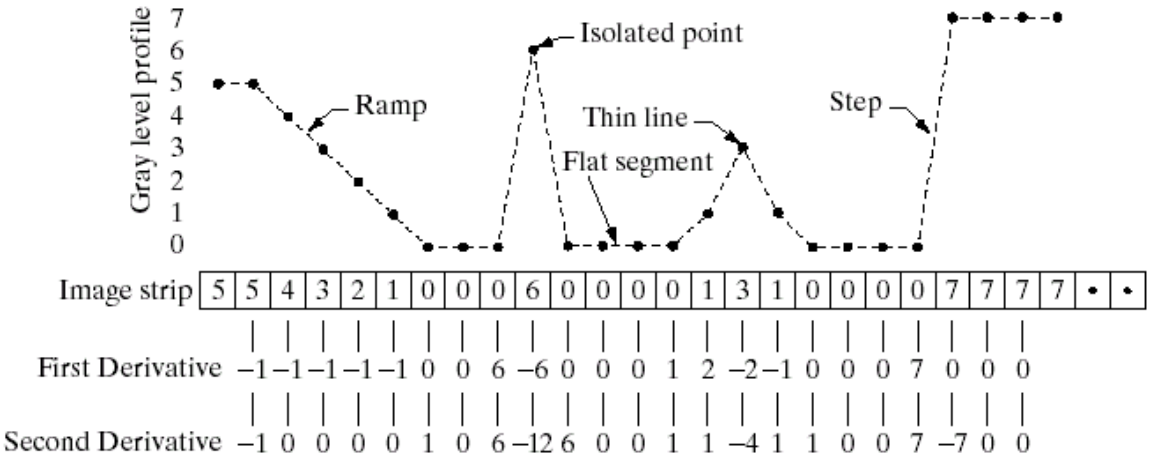
Sharpening Spatial Filters
An Example

a b
c

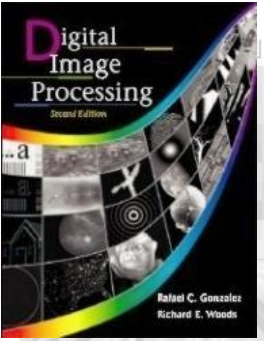
FIGURE 3.38
(a) A simple image. (b) 1-D horizontal gray-level profile along the center of the image and including the isolated noise point.
(c) Simplified profile (the points are joined by dashed lines to simplify interpretation).







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Use of Second Derivatives for Enhancement The Laplacian

- Development of the Laplacian method
  - The two dimensional Laplacian operator for continuous functions:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

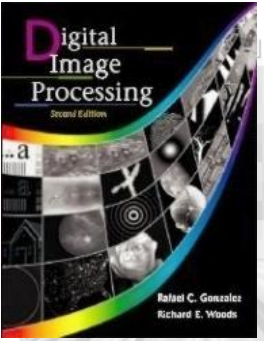
  - The Laplacian is a linear operator.

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\Rightarrow \nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x)$$

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Use of Second Derivatives for Enhancement
The Laplacian

0	1	0	1	1	1
1	−4	1	1	−8	1
0	1	0	1	1	1

0	−1	0	−1	−1	−1
−1	4	−1	−1	8	−1
0	−1	0	−1	−1	−1

a b
c d

FIGURE 3.39
(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4). (b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

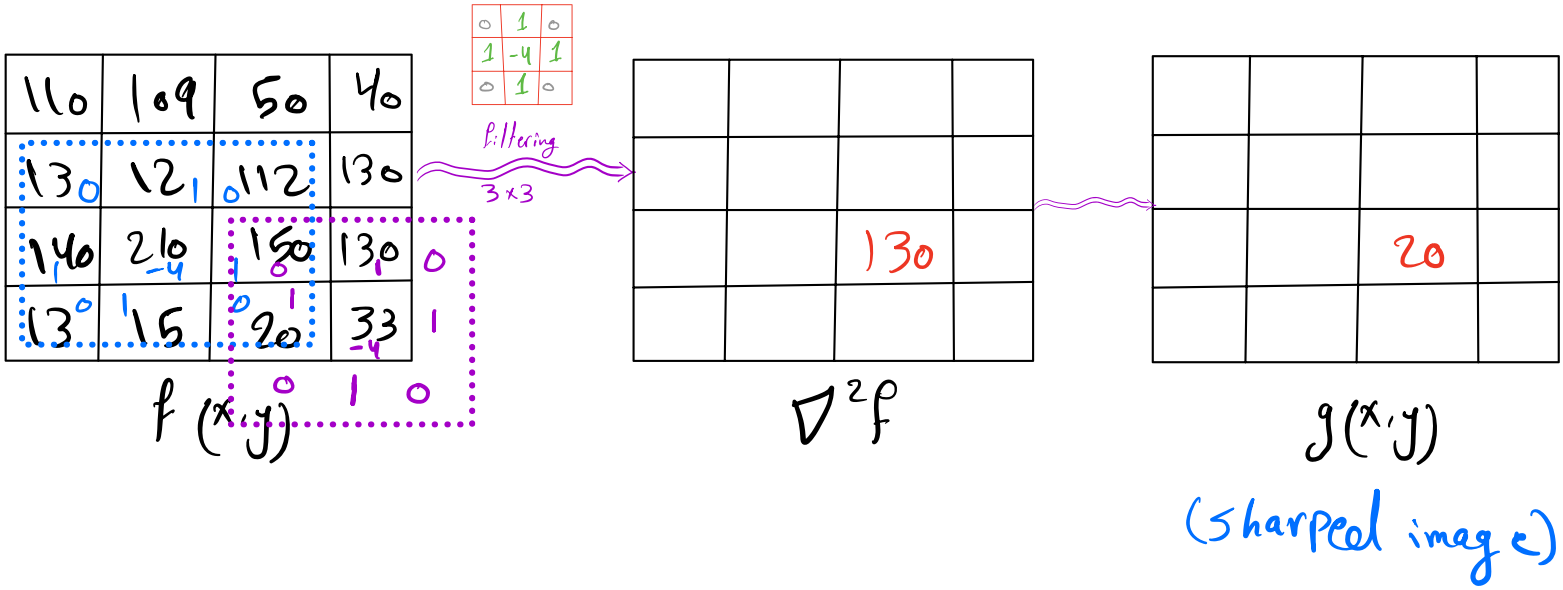
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$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$= f(x+1,y) + f(x-1,y) - 2f(x,y) + f(x,y-1) + f(x,y+1) - 2f(x,y)$$

$$= 1*f(x+1,y) + 1*f(x-1,y) + 1*f(x,y-1) + 1*f(x,y+1) - 4f(x,y)$$

0	1	0
1	-4	1
0	1	0



(sharpened image)

laplacian

40	50	100	110
40	50	100	110
40	50	100	110
40	50	100	110

$f(x,y)$

70	10	140	
	-40		

$\nabla^2 f$

110	60		

$g(x,y) = \nabla^2 f + f(x,y)$

40	50	100	110
40	50	100	110
40	50	100	110
40	50	100	110

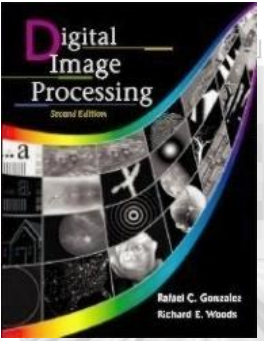
$f(x,y)$

composite laplacian

0	-1	0
-1	5	-1
0	-1	0

110	60		

$g(x,y)$



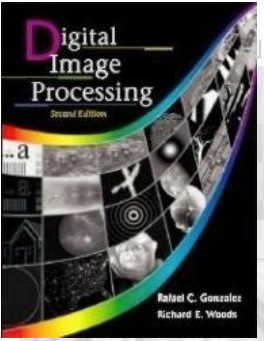
Use of Second Derivatives for Enhancement The Laplacian

- To sharpen an image, the Laplacian of the image is subtracted from the original image.

$$g(x,y)=\begin{cases} f(x,y)-\nabla^2 f & \text{if the center coefficient of the Laplacian mask is negative.} \\ f(x,y)+\nabla^2 f & \text{if the center coefficient of the Laplacian mask is positive.} \end{cases}$$

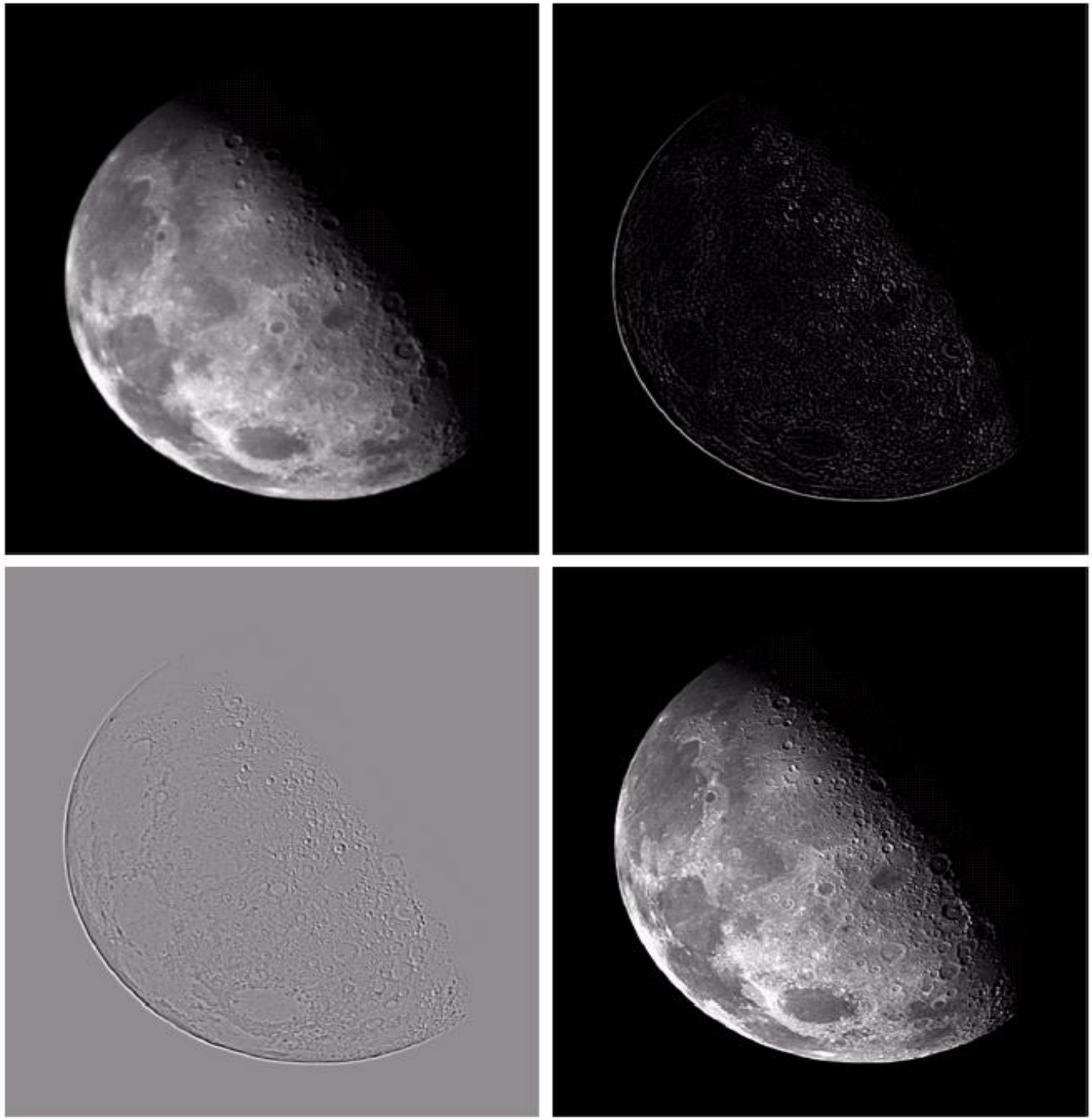
- Example: Figure 3.40

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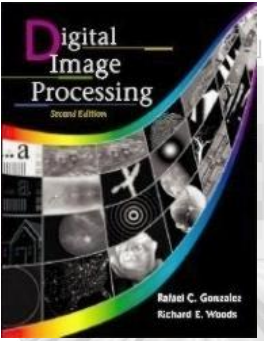


a b
c d

FIGURE 3.40
(a) Image of the North Pole of the moon.
(b) Laplacian-filtered image.
(c) Laplacian image scaled for display purposes.
(d) Image enhanced by using Eq. (3.7-5).
(Original image courtesy of NASA.)



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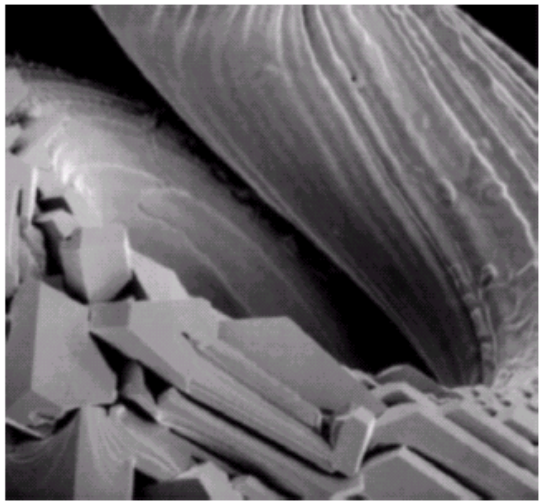
Use of Second Derivatives for Enhancement
The Laplacian: Simplifications

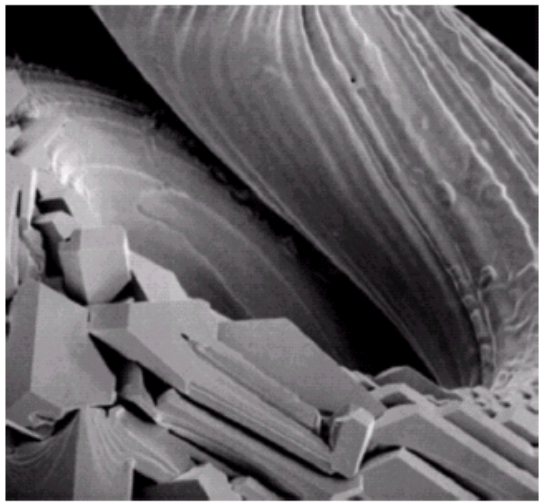
0	−1	0
−1	5	−1
0	−1	0

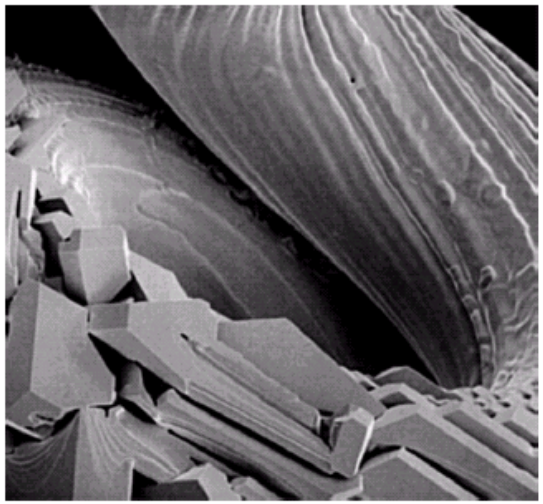
−1	−1	−1
−1	9	−1
−1	−1	−1

The $g(x,y)$ mask

Not only $\nabla^2 f$







a b c
d e

FIGURE 3.41 (a) Composite Laplacian mask. (b) A second composite mask. (c) Scanning electron microscope image. (d) and (e) Results of filtering with the masks in (a) and (b), respectively. Note how much sharper (e) is than (d). (Original image courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene.)

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effect of the Laplacian simply by adding the Laplacian image to the original. As noted in the previous paragraph, it is important to keep in mind which definition of the Laplacian is used. If the definition used has a negative center coefficient, then we *subtract*, rather than add, the Laplacian image to obtain a sharpened result. Thus, the basic way in which we use the Laplacian for image sharpening is

$$g(x, y) = f(x, y) + c\left[\nabla^2 f(x, y)\right] \tag{3.6-7}$$

where $f(x, y)$ and $g(x, y)$ are the input and sharpened images, respectively. The constant is $c = -1$ if the Laplacian filters in Fig. 3.37(a) or (b) are used, and $c = 1$ if either of the other two filters is used.

EXAMPLE 3.15: Image sharpening using the Laplacian.

■ Figure 3.38(a) shows a slightly blurred image of the North Pole of the moon. Figure 3.38(b) shows the result of filtering this image with the Laplacian mask in Fig. 3.37(a). Large sections of this image are black because the Laplacian contains both positive and negative values, and all negative values are clipped at 0 by the display.

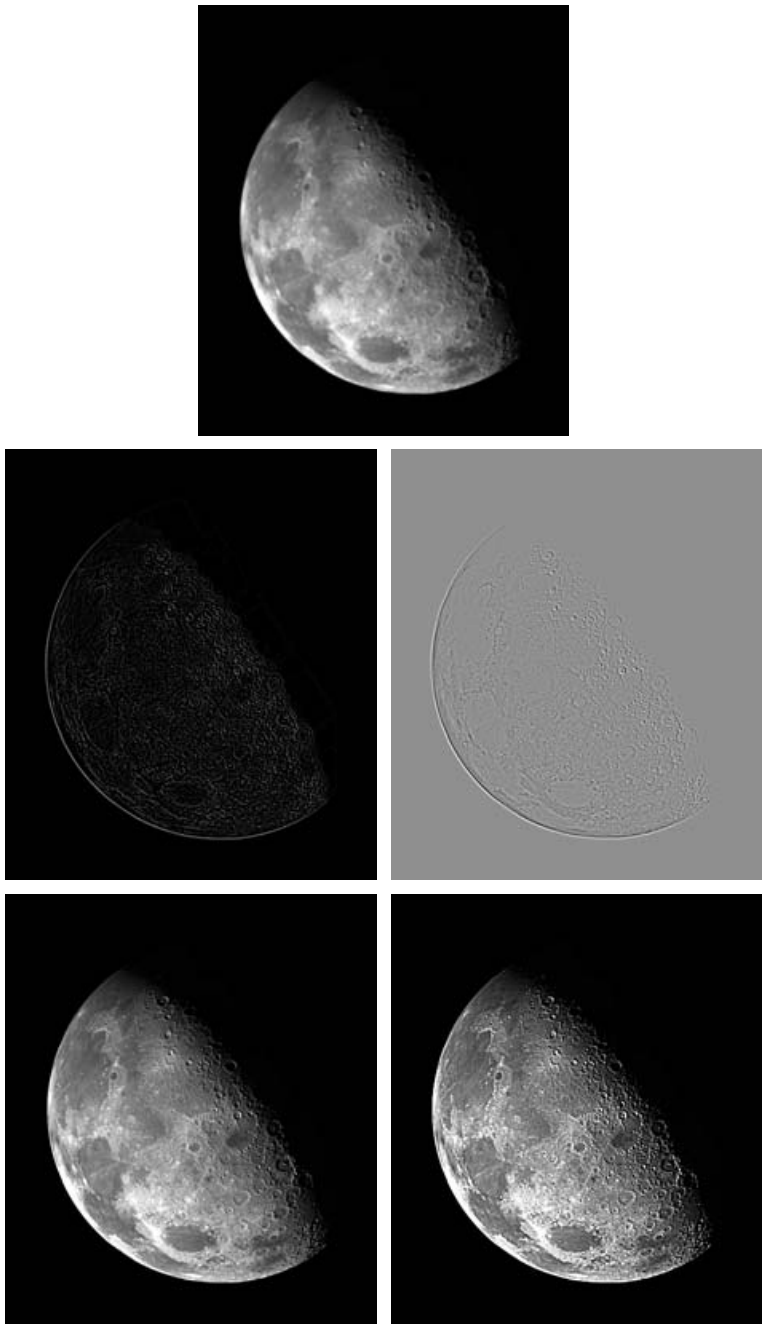
A typical way to scale a Laplacian image is to add to it its minimum value to bring the new minimum to zero and then scale the result to the full $[0, L - 1]$ intensity range, as explained in Eqs. (2.6-10) and (2.6-11). The image in Fig. 3.38(c) was scaled in this manner. Note that the dominant features of the image are edges and sharp intensity discontinuities. The background, previously black, is now gray due to scaling. This grayish appearance is typical of Laplacian images that have been scaled properly. Figure 3.38(d) shows the result obtained using Eq. (3.6-7) with $c = -1$. The detail in this image is unmistakably clearer and sharper than in the original image. Adding the original image to the Laplacian restored the overall intensity variations in the image, with the Laplacian increasing the contrast at the locations of intensity discontinuities. The net result is an image in which small details were enhanced and the background tonality was reasonably preserved. Finally, Fig. 3.38(e) shows the result of repeating the preceding procedure with the filter in Fig. 3.37(b). Here, we note a significant improvement in sharpness over Fig. 3.38(d). This is not unexpected because using the filter in Fig. 3.37(b) provides additional differentiation (sharpening) in the diagonal directions. Results such as those in Figs. 3.38(d) and (e) have made the Laplacian a tool of choice for sharpening digital images. ■

3.6.3 Unsharp Masking and Highboost Filtering

A process that has been used for many years by the printing and publishing industry to sharpen images consists of subtracting an unsharp (smoothed) version of an image from the original image. This process, called *unsharp masking*, consists of the following steps:

AVG. filter أي باستخدام ←

1. Blur the original image.
2. Subtract the blurred image from the original (the resulting difference is called the *mask*.)
3. Add the mask to the original.



a
b c
d e

FIGURE 3.38
(a) Blurred image of the North Pole of the moon. (b) Laplacian without scaling. (c) Laplacian with scaling. (d) Image sharpened using the mask in Fig. 3.37(a). (e) Result of using the mask in Fig. 3.37(b). (Original image courtesy of NASA.)

Letting $\overline{f}(x, y)$ denote the blurred image, unsharp masking is expressed in equation form as follows. First we obtain the mask:

$$g_{\text{mask}}(x, y) = f(x, y) - \overline{f}(x, y) \tag{3.6-8}$$

Then we add a weighted portion of the mask back to the original image:

$$g(x, y) = f(x, y) + k * g_{\text{mask}}(x, y) \tag{3.6-9}$$

where we included a weight, k ($k \geq 0$), for generality. When $k = 1$, we have unsharp masking, as defined above. When $k > 1$, the process is referred to as

a
b
c
d

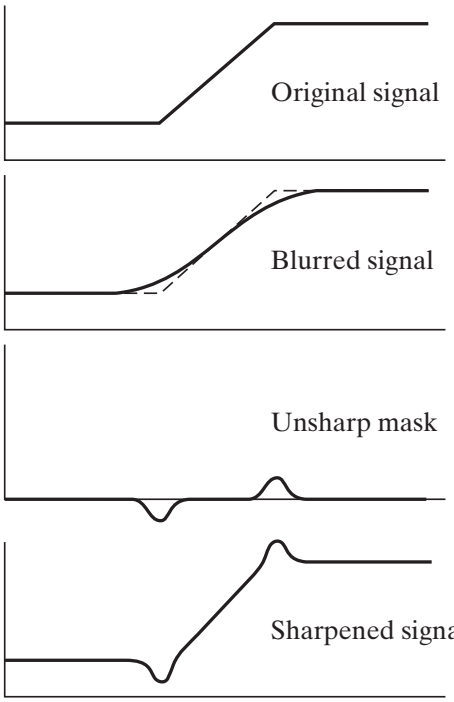


FIGURE 3.39 1-D illustration of the mechanics of unsharp masking. (a) Original signal. (b) Blurred signal with original shown dashed for reference. (c) Unsharp mask. (d) Sharpened signal, obtained by adding (c) to (a).

highboost filtering. Choosing $k < 1$ de-emphasizes the contribution of the unsharp mask.

Figure 3.39 explains how unsharp masking works. The intensity profile in Fig. 3.39(a) can be interpreted as a horizontal scan line through a vertical edge that transitions from a dark to a light region in an image. Figure 3.39(b) shows the result of smoothing, superimposed on the original signal (shown dashed) for reference. Figure 3.39(c) is the unsharp mask, obtained by subtracting the blurred signal from the original. By comparing this result with the section of Fig. 3.36(c) corresponding to the ramp in Fig. 3.36(a), we note that the unsharp mask in Fig. 3.39(c) is very similar to what we would obtain using a second-order derivative. Figure 3.39(d) is the final sharpened result, obtained by adding the mask to the original signal. The points at which a change of slope in the intensity occurs in the signal are now emphasized (sharpened). Observe that negative values were added to the original. Thus, it is possible for the final result to have negative intensities if the original image has any zero values or if the value of k is chosen large enough to emphasize the peaks of the mask to a level larger than the minimum value in the original. Negative values would cause a dark halo around edges, which, if k is large enough, can produce objectionable results.

EXAMPLE 3.16: Image sharpening using unsharp masking.

■ Figure 3.40(a) shows a slightly blurred image of white text on a dark gray background. Figure 3.40(b) was obtained using a Gaussian smoothing filter (see Section 3.4.4) of size 5×5 with $\sigma = 3$. Figure 3.40(c) is the unsharp mask, obtained using Eq. (3.6-8). Figure 3.40(d) was obtained using unsharp



a
b
c
d
e

FIGURE 3.40 (a) Original image. (b) Result of blurring with a Gaussian filter. (c) Unsharp mask. (d) Result of using unsharp masking. (e) Result of using highboost filtering.

masking [Eq. (3.6-9) with $k = 1$]. This image is a slight improvement over the original, but we can do better. Figure 3.40(e) shows the result of using Eq. (3.6-9) with $k = 4.5$, the largest possible value we could use and still keep positive all the values in the final result. The improvement in this image over the original is significant. ■

3.6.4 Using First-Order Derivatives for (Nonlinear) Image Sharpening—The Gradient

First derivatives in image processing are implemented using the magnitude of the gradient. For a function $f(x, y)$, the gradient of f at coordinates (x, y) is defined as the two-dimensional column *vector*

$$\nabla f \equiv \text{grad}(f) \equiv \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix} \tag{3.6-10}$$

We discuss the gradient in detail in Section 10.2.5. Here, we are interested only in using the magnitude of the gradient for image sharpening.

This vector has the important geometrical property that it points in the direction of the greatest rate of change of f at location (x, y).

The *magnitude* (*length*) of vector ∇f, denoted as $M(x, y)$, where

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2} \tag{3.6-11}$$

is the *value* at (x, y) of the rate of change in the direction of the gradient vector. Note that $M(x, y)$ is an image of the same size as the original, created when x and y are allowed to vary over all pixel locations in f. It is common practice to refer to this image as the *gradient image* (or simply as the *gradient* when the meaning is clear).

un sharp Masking :

50	60	70	80
30	20	30	20
50	60	70	80
100	100	100	100

f(x,y)

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

standard Avg. mask

		54	

b = blured image

		-24	

mask = f(x,y) - b

(blur mask)

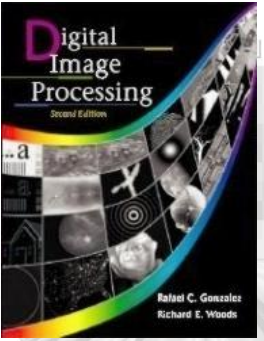
↳ with high boost

بعزب كل خلية بالرقم

لتكبير كل القيم

		6	

result = f(x,y) + mask



Use of First Derivatives for Enhancement
The Gradient

Edge detection

- Development of the Gradient method
  - The gradient of function f at coordinates (x,y) is defined as the two-dimensional column vector:

$$\nabla \mathbf{f} = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

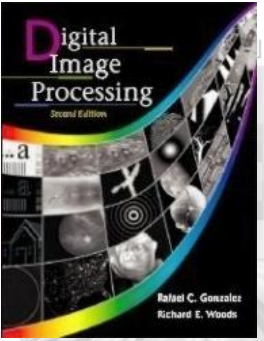
  - The magnitude of this vector is given by

$$\nabla f = \text{mag}(\nabla \mathbf{f}) = \left[G_x^2 + G_y^2\right]^{\frac{1}{2}} = \left[\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2\right]^{\frac{1}{2}}$$

Hori. Ver.

$$\nabla f \approx |G_x| + |G_y|$$

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Use of First Derivatives for Enhancement
The Gradient

a
b c
d e

FIGURE 3.44
A 3 × 3 region of an image (the z's are gray-level values) and masks used to compute the gradient at point labeled z_5. All masks coefficients sum to zero, as expected of a derivative operator.

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

−1	0
0	1

0	−1
1	0

Roberts cross-gradient operators

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

Sobel operators

$g_{(x)}$ $g_{(y)}$

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40	50	100	110
40	50	100	110
40	50	100	110
40	50	100	110
40	50	100	110
40	50	100	110

$f(x,y)$

G_x

-1	-2	-1
0	0	0
1	2	1

$G(x)$

	0	0	

$\nabla f \approx |G_x| + |G_y|$

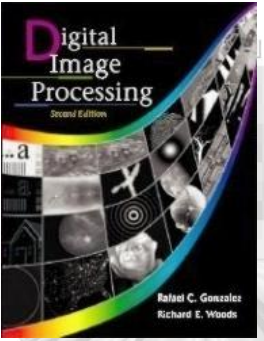
	240	240	

G_y

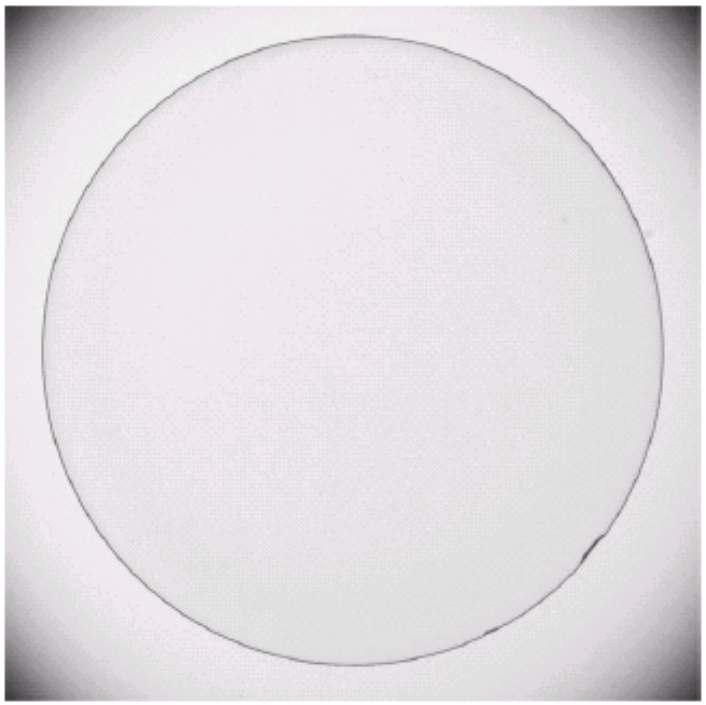
-1	0	1
-2	0	2
-1	0	1

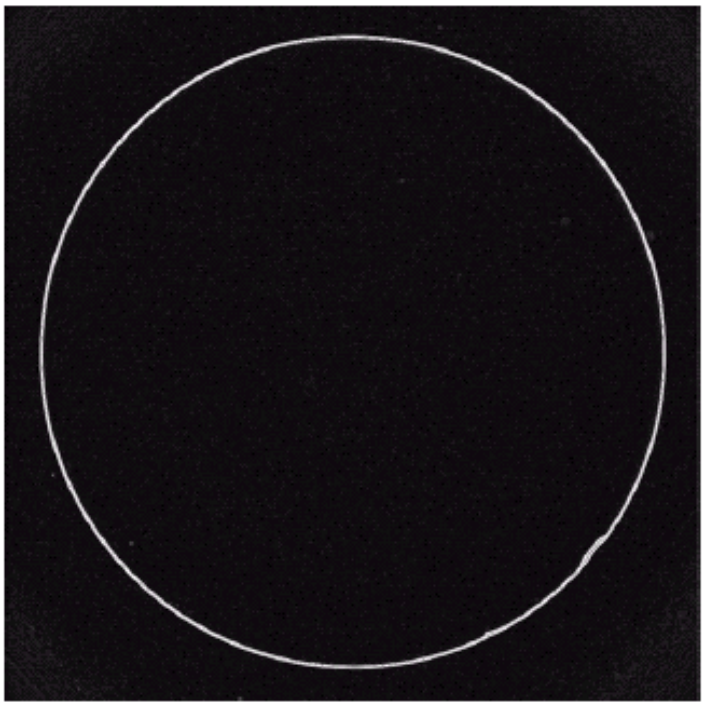
$G(y)$

	240	240	



Use of First Derivatives for Enhancement
The Gradient: Using Sobel Operators

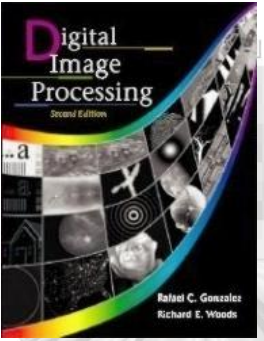




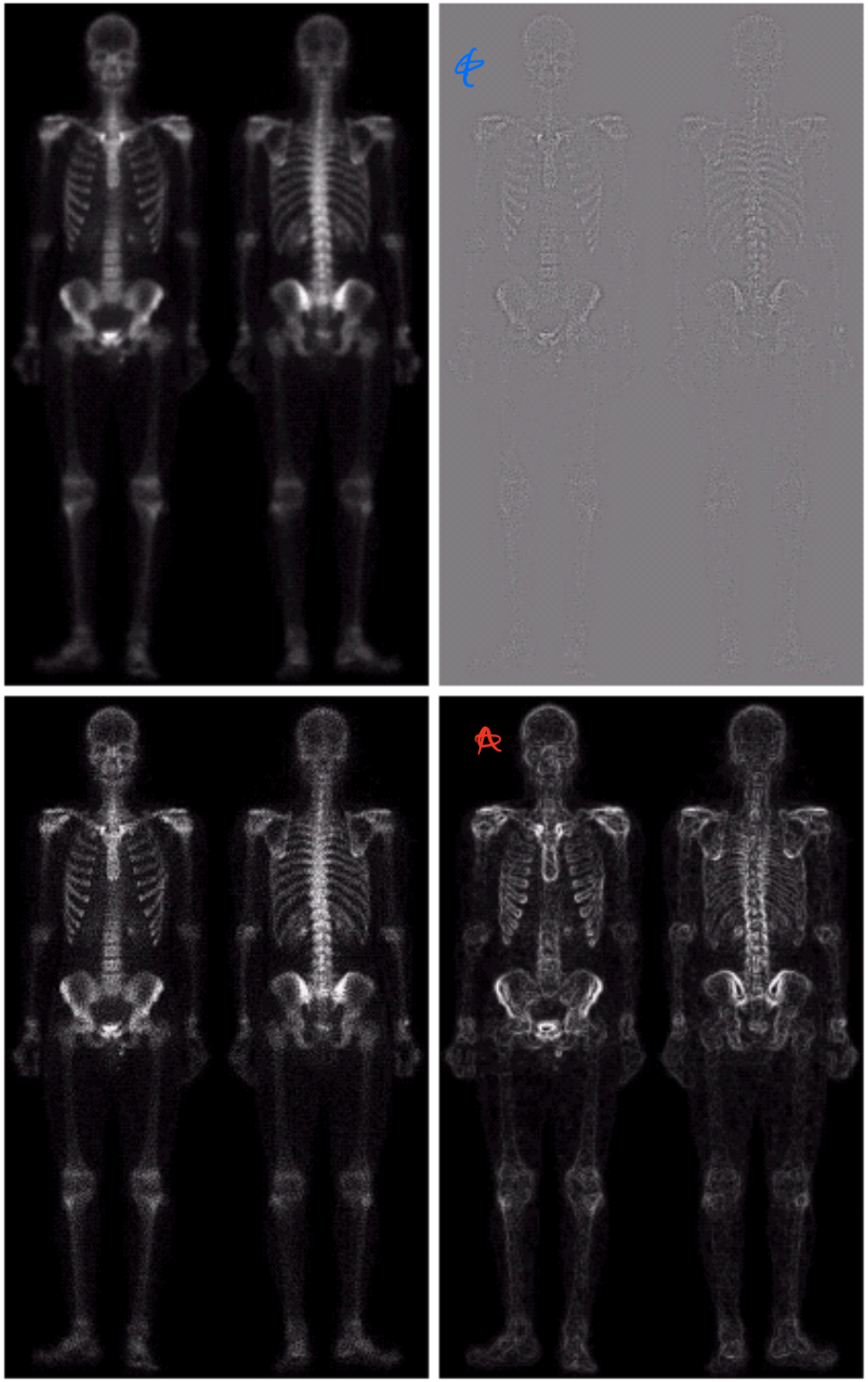
a b

FIGURE 3.45
Optical image of contact lens (note defects on the boundary at 4 and 5 o'clock). (b) Sobel gradient. (Original image courtesy of Mr. Pete Sites, Perceptics Corporation.)

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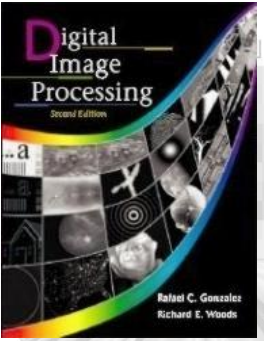
Combining Spatial Enhancement Methods



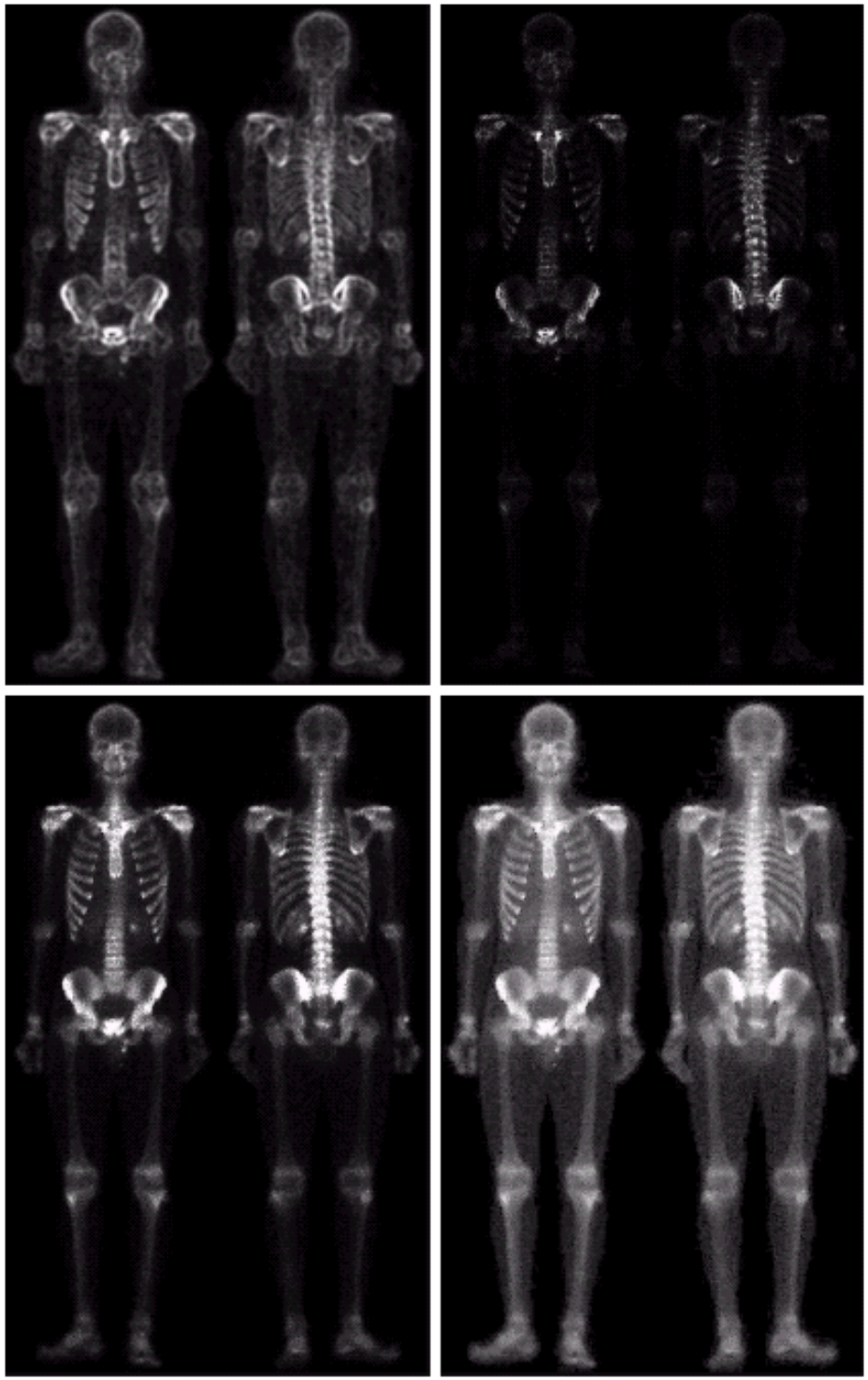
a b
c d

FIGURE 3.46
(a) Image of whole body bone scan. (b) Laplacian of (a). (c) Sharpened image obtained by adding (a) and (b). (d) Sobel of (a).

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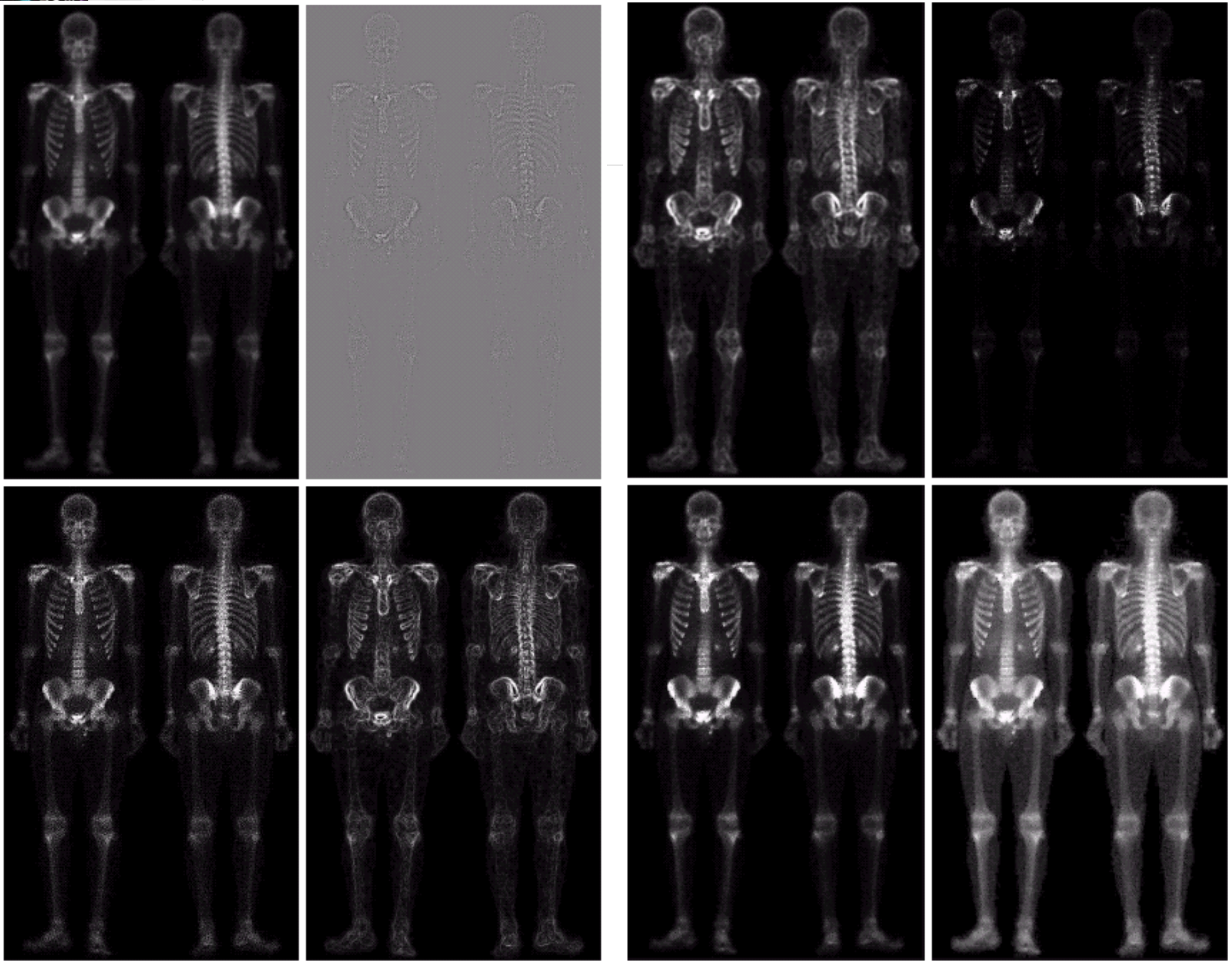
Combining Spatial Enhancement Methods



e f
g h

FIGURE 3.46
(Continued)
(e) Sobel image smoothed with a 5 × 5 averaging filter. (f) Mask image formed by the product of (c) and (e). (g) Sharpened image obtained by the sum of (a) and (f). (h) Final result obtained by applying a power-law transformation to (g). Compare (g) and (h) with (a). (Original image courtesy of G.E. Medical Systems.)

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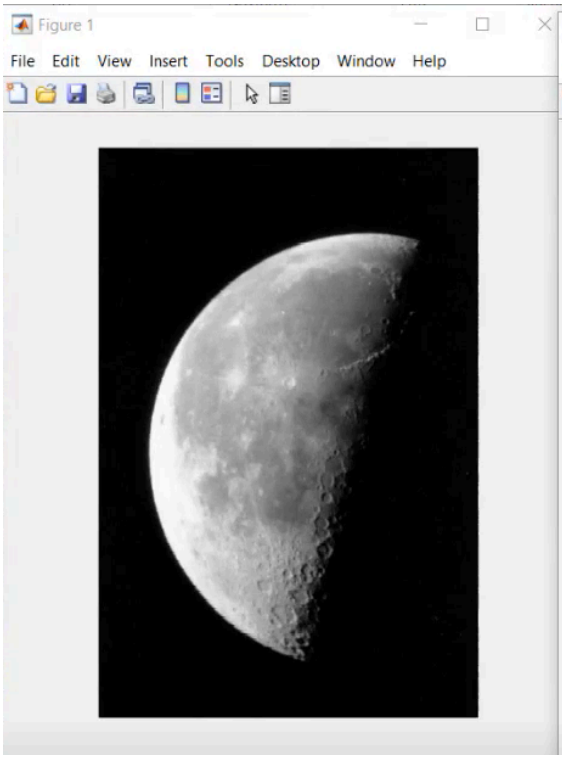
```
i=imread('moon.tif');
w1=fspecial('laplacian',0);
w2=[1 1 1;1 -8 1;1 1 1];
im1=i-imfilter(i,w1);
im2=i-imfilter(i,w2);
imshow(i);figure;imshow(im1);figure;imshow(im2);
```

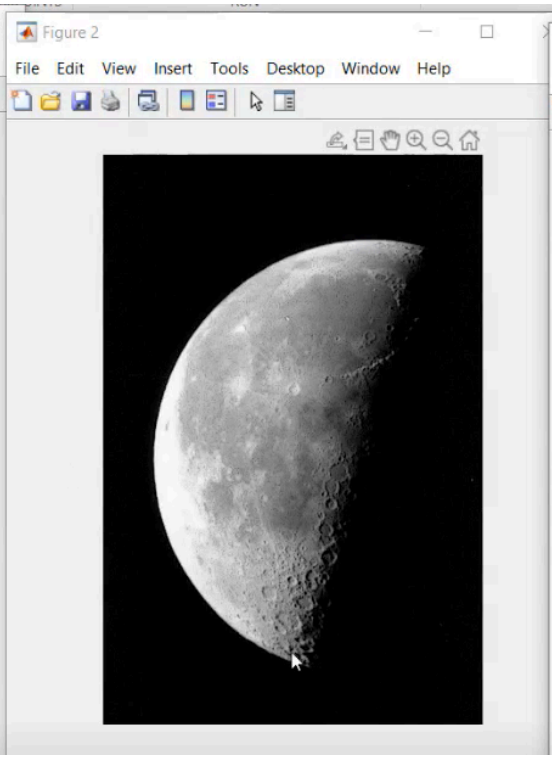
→ center = -4

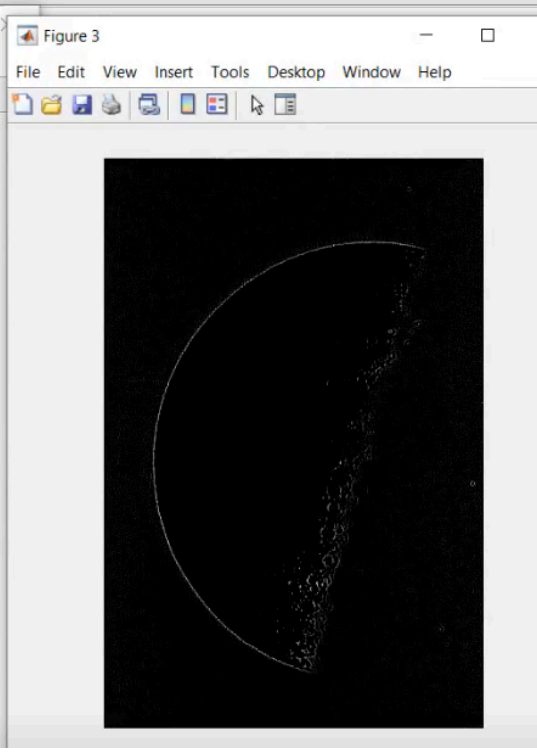
→ center = -8

هي حال كنت بدي استعمل فلتر غير اي بشترط تجمعه = 0 لازم اوطي القيم بمتر كما بعدها رفي به

0	1	0	1	1	1
1	−4	1	1	−8	1
0	1	0	1	1	1



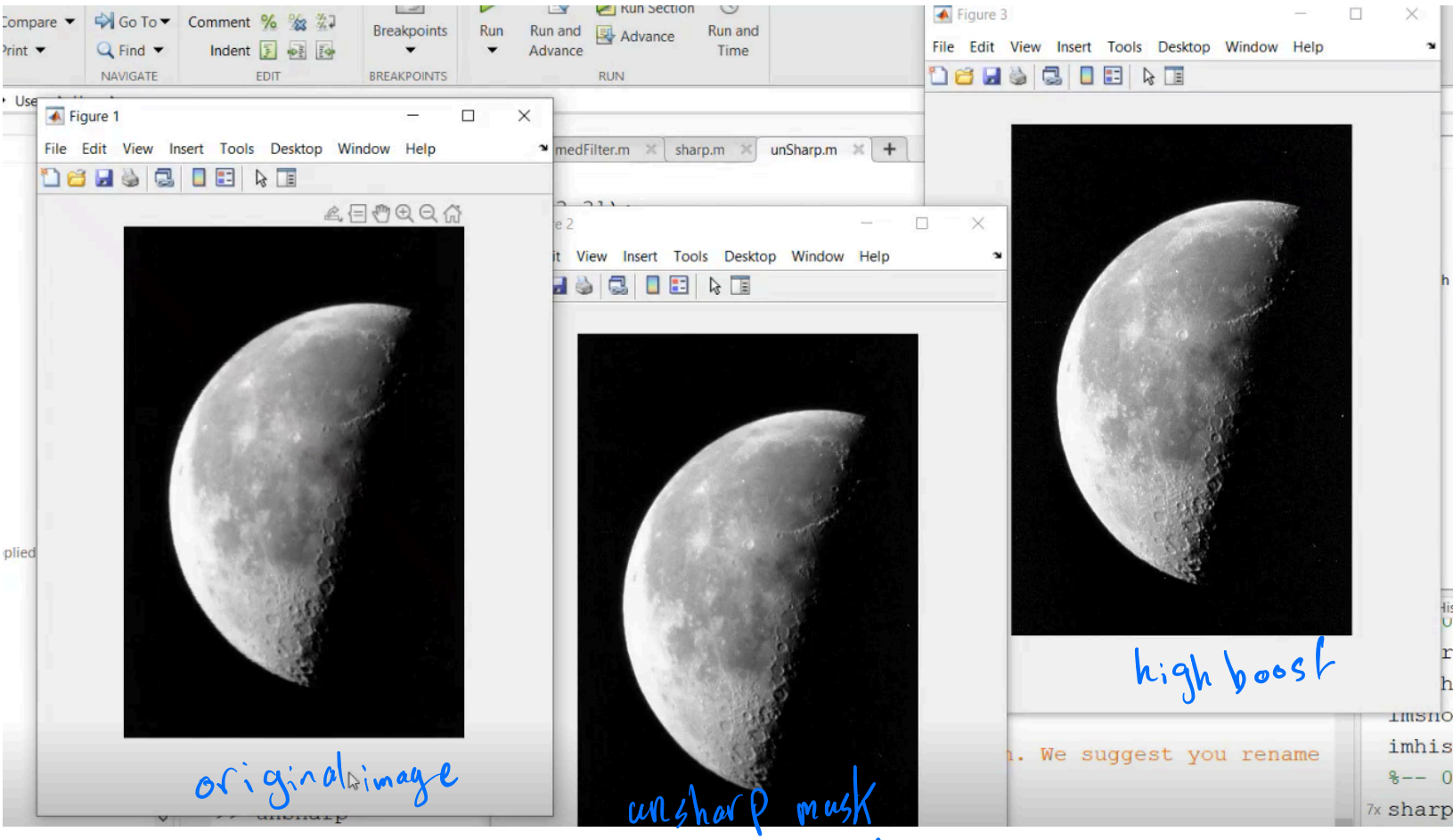




example for unsharping mask for sharpening

```
i=imread('moon.tif');
w=fspecial('average',[3 3]);
j=imfilter(i,w);
unMask=i-j;
sharp=i+unMask;
sharpHigh=i+unMask*3;
imshow(i);figure;imshow(sharp);figure;imshow(sharpHigh);
```

→ for high boost filter



for sharping filter

```
i=imread('cameraman.tif');
h=fspecial('sobel');
v=h';
jH=imfilter(i,h);
jV=imfilter(i,v);
jFinal=abs(jH)+abs(jV);
imshow(jFinal);
```

horizontal ← default

sobel filter

vertical

h transpose



```
i=imread('cameraman.tif');
h=fspecial('sobel');
v=h';
jH=imfilter(i,h);
jV=imfilter(i,v);
jFinal=abs(jH)+abs(jV);
imshow(jH);
```

Horizantly det edges



```
i=imread('cameraman.tif');
h=fspecial('sobel');
v=h';
jH=imfilter(i,h);
jV=imfilter(i,v);
jFinal=abs(jH)+abs(jV);
imshow(jV);
```

Virticaly edges

